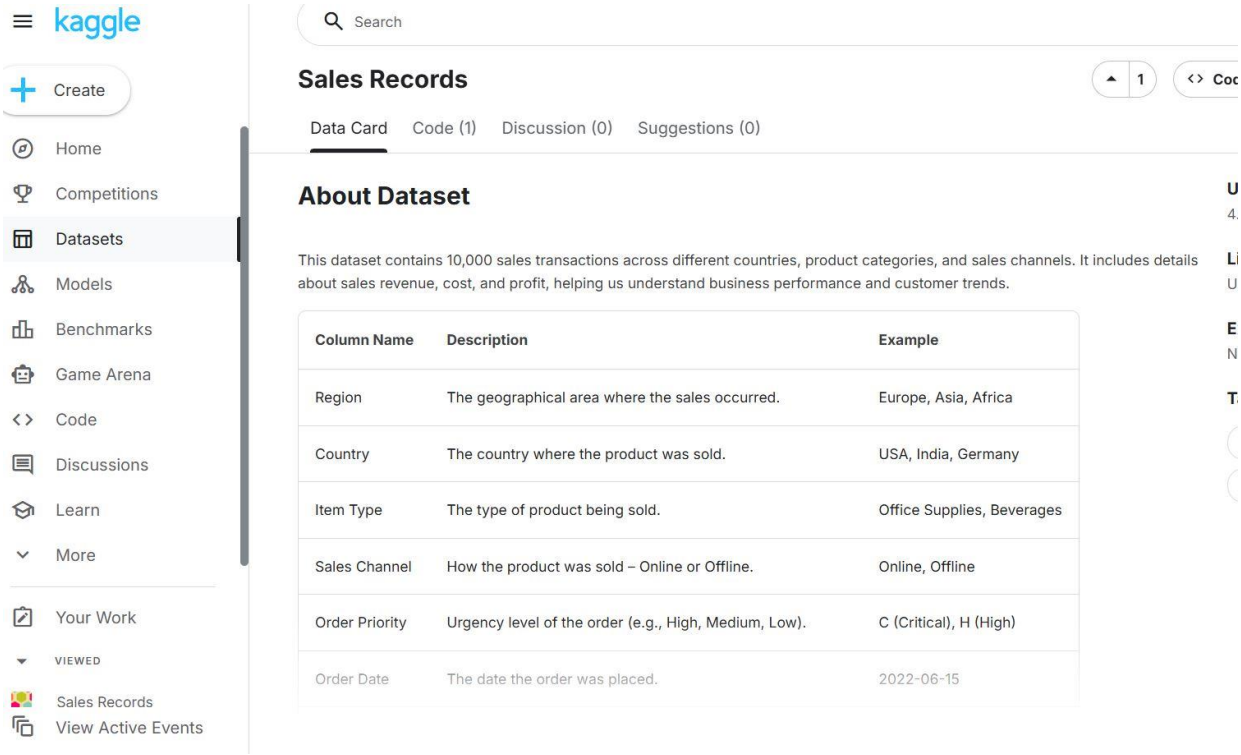
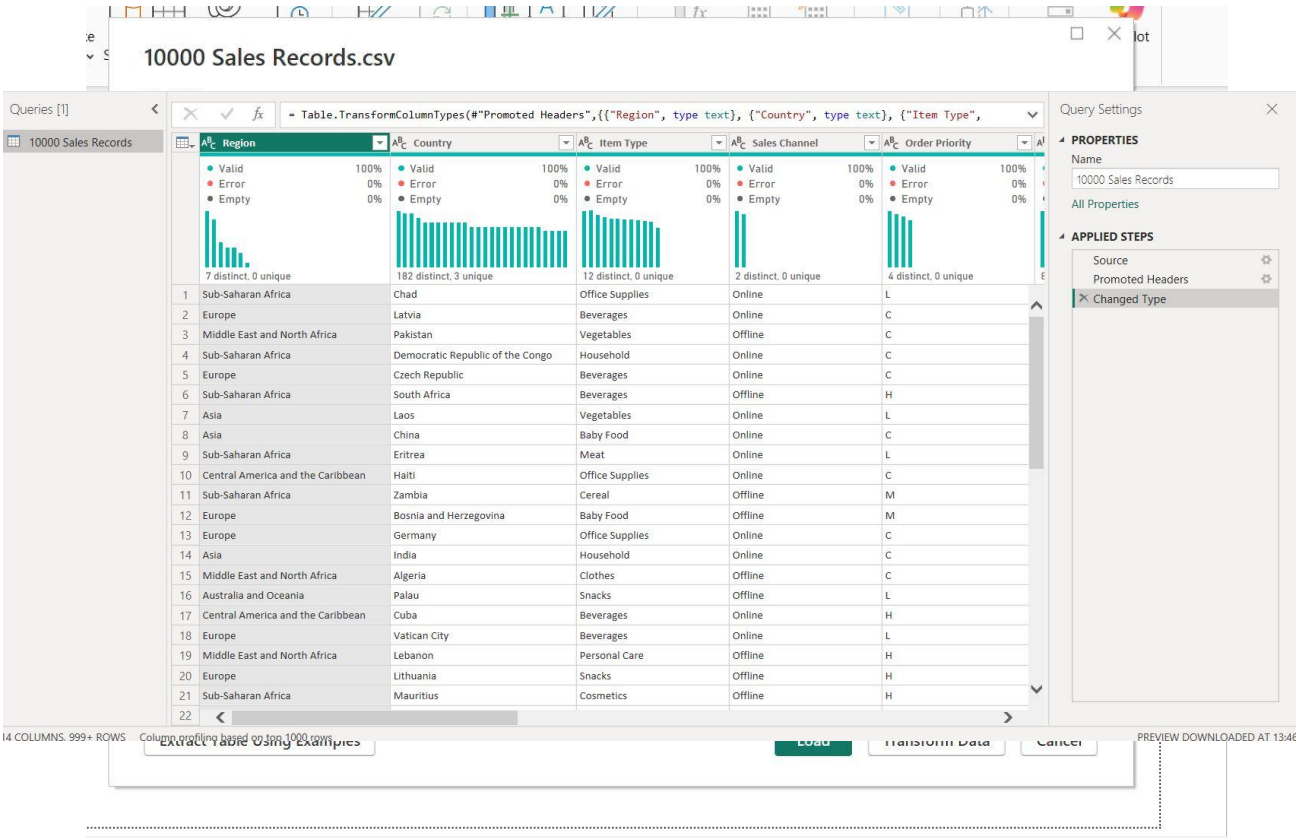
PETER KIDIGA -671341

Section 1 - Process Evidence

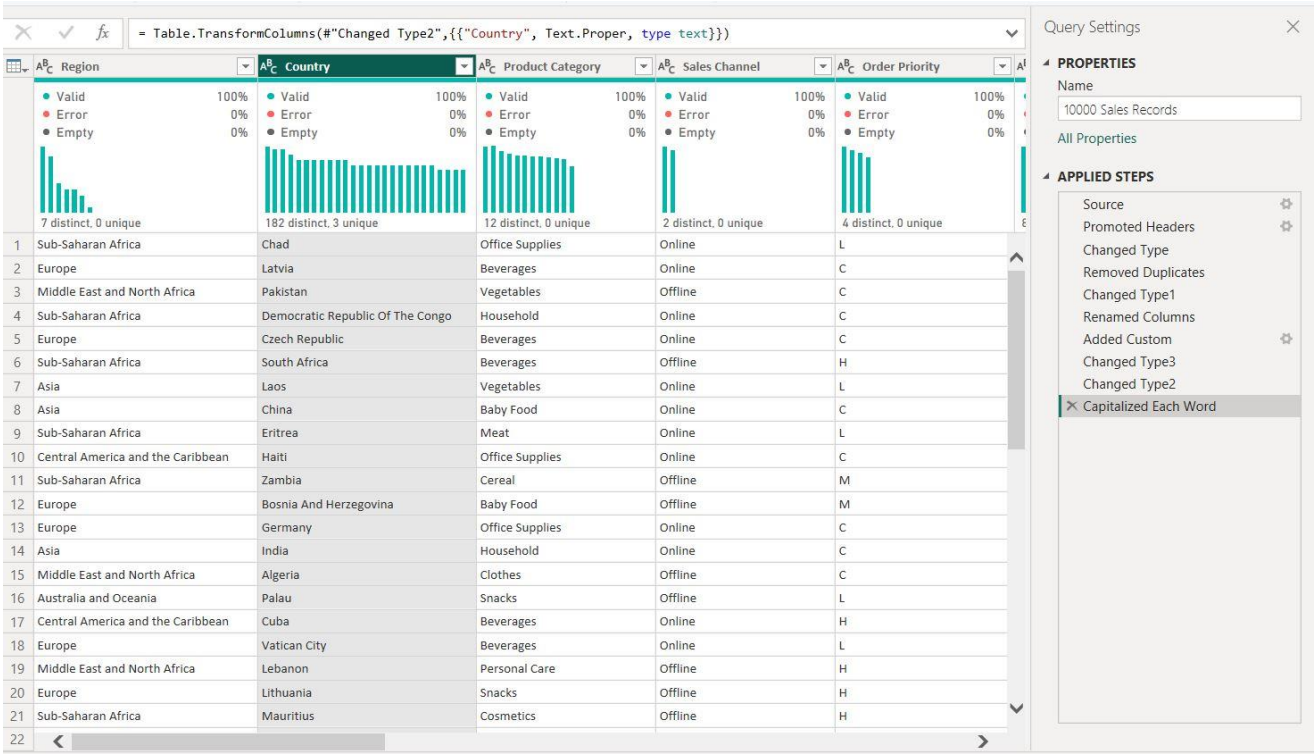


Evidence of dataset sourcing from Kaggle. The selected dataset contains exactly 10,000 sales transactions, meeting the exam's requirement of a minimum of 7,000 rows. It includes essential attributes like Region, Item Type, and Order Date for analytical storytelling



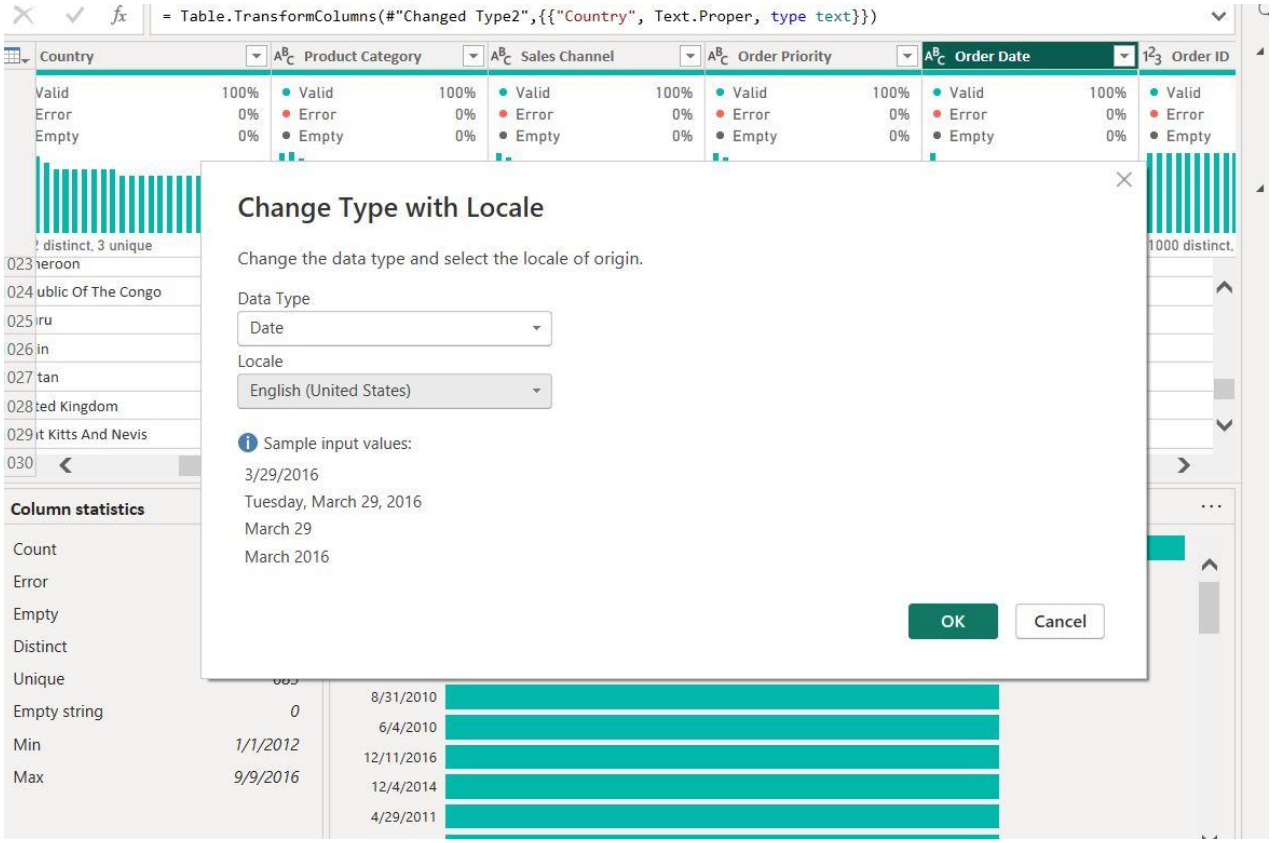
Initial data loading process. This screenshot shows the Power BI Navigator previewing the 10,000-row sales dataset, confirming a successful connection to the source file.

Accessing the Power Query Editor. This environment will be used to perform the 10+ required transformation steps, such as data type correction and date extraction, to prepare the data for modeling.

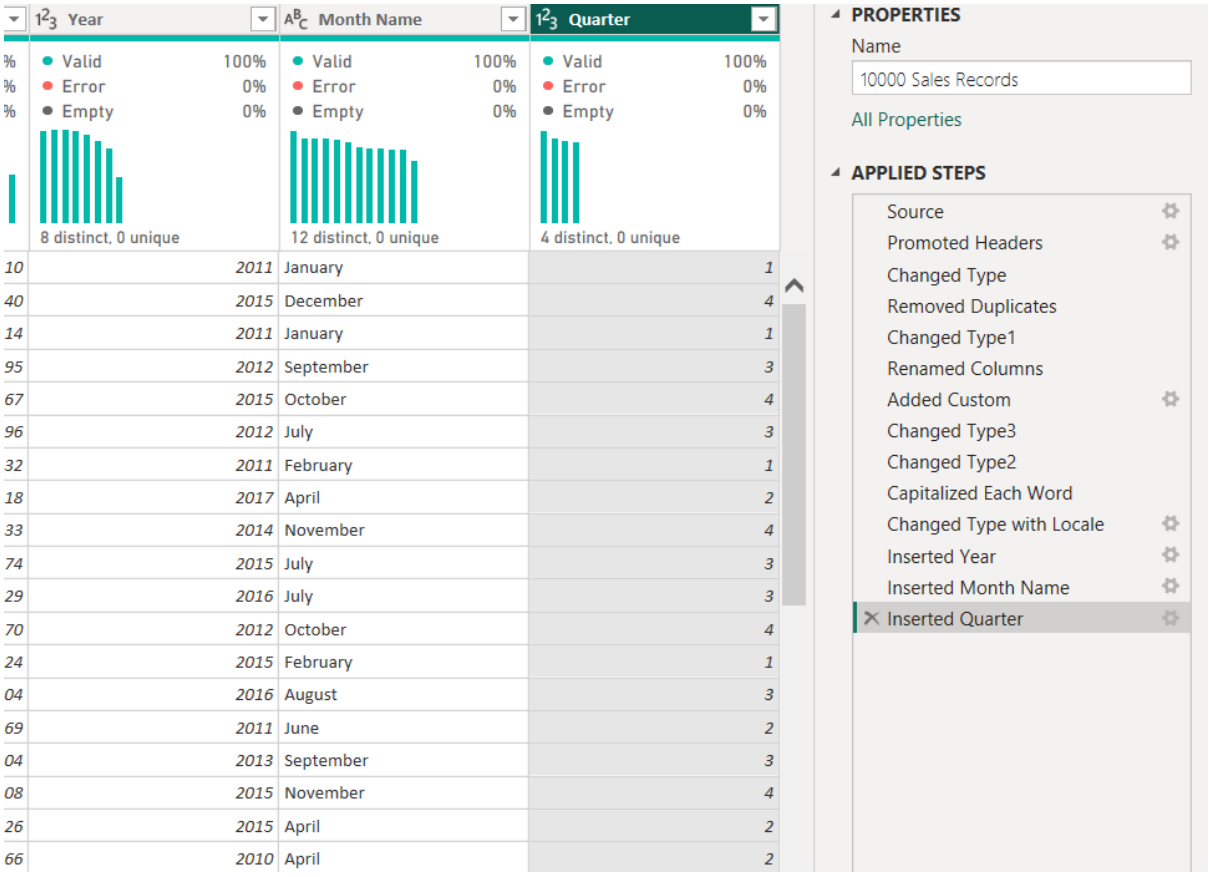


	Region	Country	Product Category	Sales Channel	Order Priority
1	Sub-Saharan Africa	Chad	Office Supplies	Online	L
2	Europe	Latvia	Beverages	Online	C
3	Middle East and North Africa	Pakistan	Vegetables	Offline	C
4	Sub-Saharan Africa	Democratic Republic Of The Congo	Household	Online	C
5	Europe	Czech Republic	Beverages	Online	C
6	Sub-Saharan Africa	South Africa	Beverages	Offline	H
7	Asia	Laos	Vegetables	Online	L
8	Asia	China	Baby Food	Online	C
9	Sub-Saharan Africa	Eritrea	Meat	Online	L
10	Central America and the Caribbean	Haiti	Office Supplies	Online	C
11	Sub-Saharan Africa	Zambia	Cereal	Offline	M
12	Europe	Bosnia And Herzegovina	Baby Food	Offline	M
13	Europe	Germany	Office Supplies	Online	C
14	Asia	India	Household	Online	C
15	Middle East and North Africa	Algeria	Clothes	Offline	C
16	Australia and Oceania	Palau	Snacks	Offline	L
17	Central America and the Caribbean	Cuba	Beverages	Online	H
18	Europe	Vatican City	Beverages	Online	L
19	Middle East and North Africa	Lebanon	Personal Care	Offline	H
20	Europe	Lithuania	Snacks	Offline	H
21	Sub-Saharan Africa	Mauritius	Cosmetics	Offline	H
22					

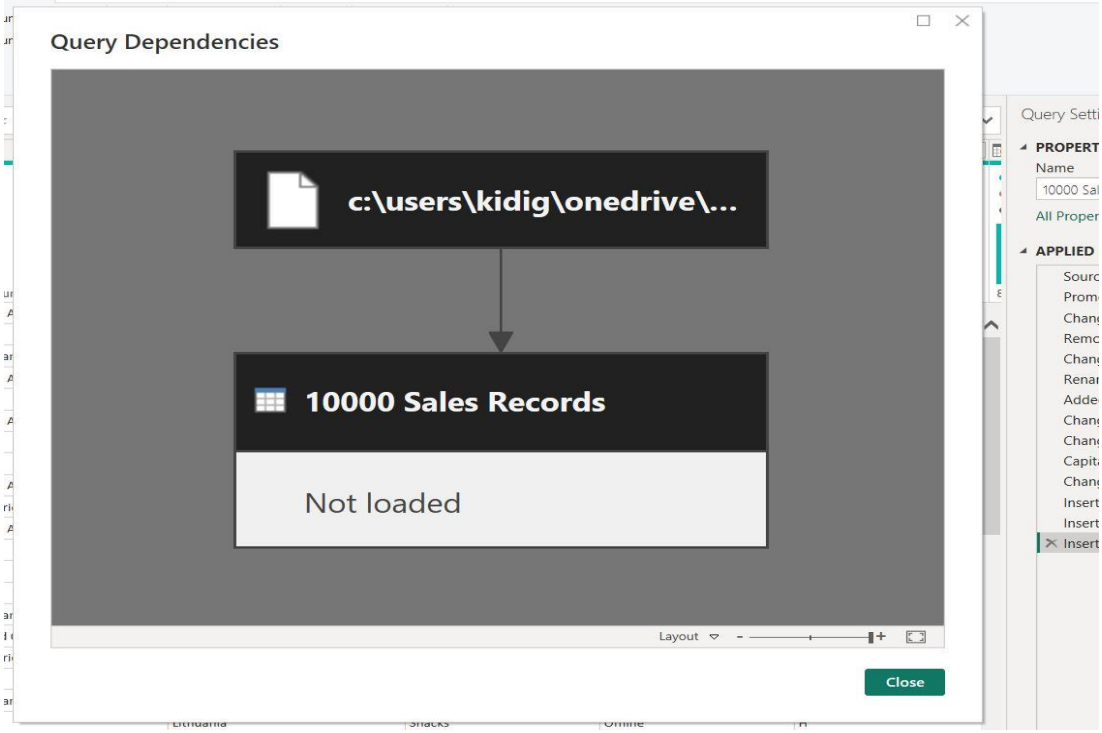
Demonstration of extensive Power Query transformations including data type correction, removal of duplicates, and the creation of a custom 'VAT' column. The Applied Steps pane confirms the systematic cleaning of the 10,000-row dataset to ensure analytical accuracy.



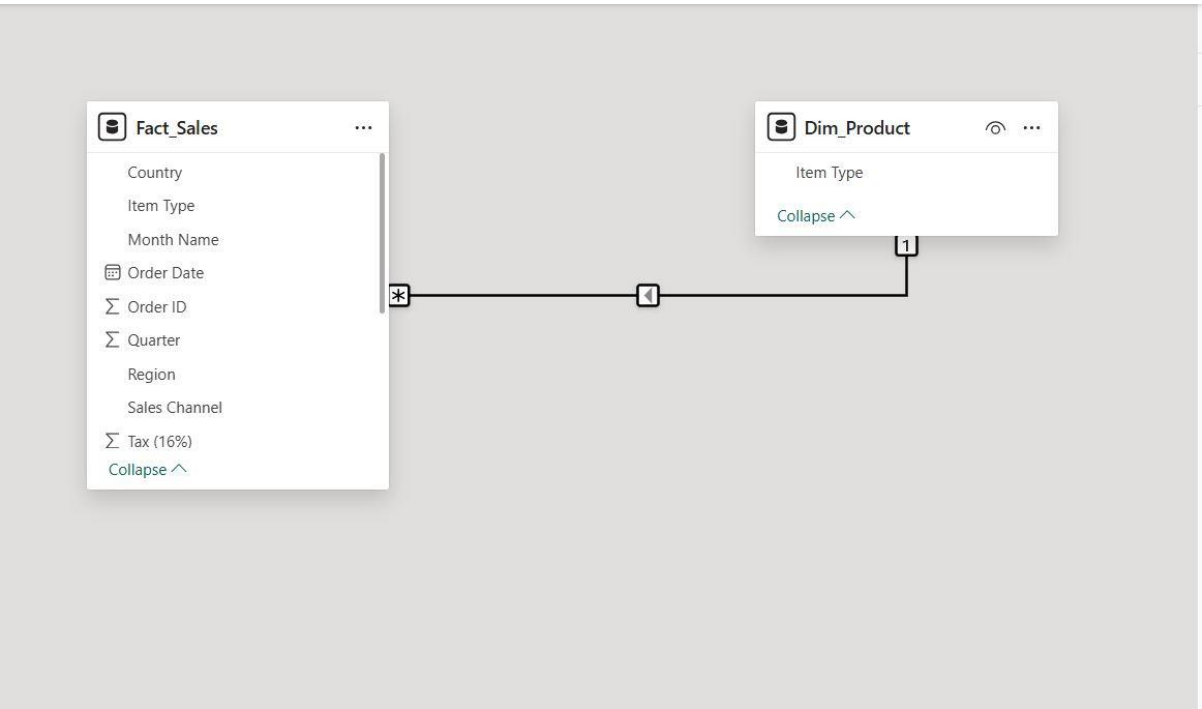
Resolving data type conversion errors using the 'Change Type with Locale' feature. This step ensures that date strings are correctly parsed according to their regional format, eliminating 'Expression.Error' values and standardizing the dataset for time-based analysis.



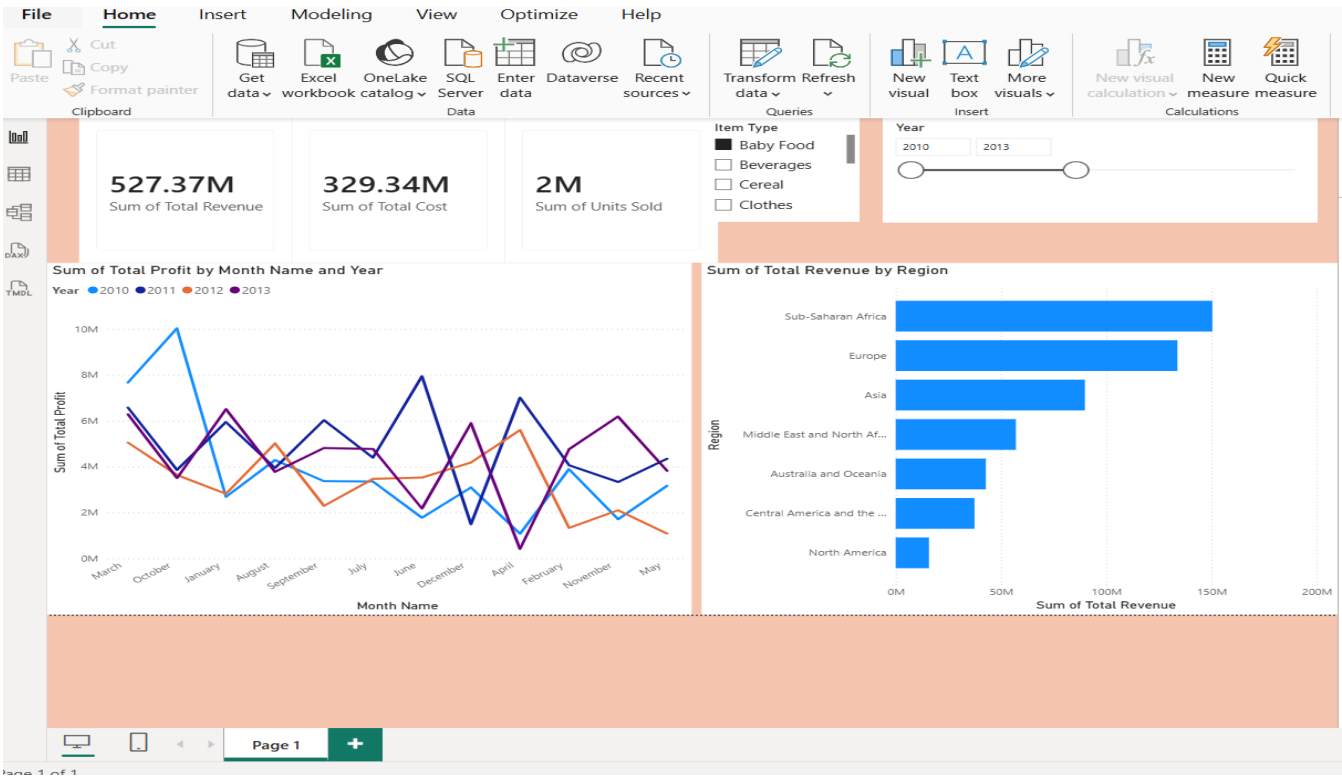
Execution of required date extractions in Power Query. Three new analytical attributes—Year, Month Name, and Quarter—were successfully derived from the primary 'Order Date' column. This ensures the dataset supports time-series analysis and granular trend reporting.



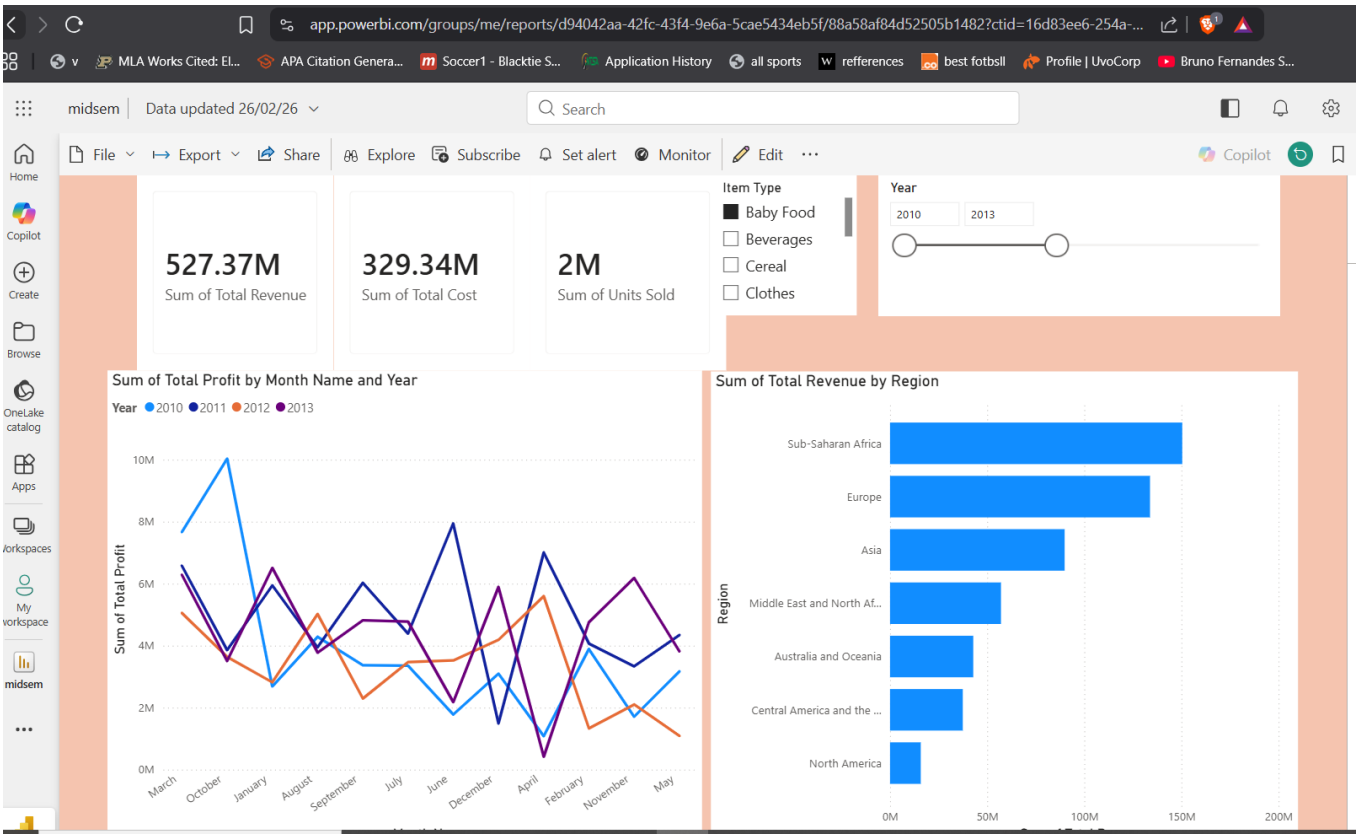
Screenshot of the Query Dependency view within Power Query. This illustrates the data lineage from the local source file on OneDrive to the '10000 Sales Records' query, confirming a documented and professional data preparation workflow.



Implementation of a Star Schema data model as required by Part B. The model consists of a central 'Fact_Sales' table and a 'Dim_Product' dimension table, connected via a One-to-Many (1:*) relationship. This structure optimizes performance and ensures analytical depth while avoiding many-to-many complexities.



Application of professional dashboard aesthetics and layout design. A custom canvas background and consistent visual formatting have been applied to enhance readability and professional appeal.



Successful publication of the Power BI report to the Power BI Service. This marks the transition from local development to cloud deployment, ensuring the solution is ready for stakeholder accessibility.